

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

```
1 public class Arithmetic{
2     public static void main(String[] args){
3         int a = 5, b= 5;
4
5         System.out.println("Addition = " + (a + b));
6         System.out.println("Subraction = " + (a - b));
7         System.out.println("MUltiplication = " + (a * b));
8         System.out.println("Division = " + (a / b));
9     }
10 }
```

OUTPUT

```
Addition = 15
Subraction = 5
MUltiplication = 50
Division = 2
```

Your INPUT goes here!

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new
6             Scanner(System.in);
7
8         int a = sc.nextInt();
9         int b = sc.nextInt();
10        int c = sc.nextInt();
11
12        if (a >= b && a >= c) {
13            System.out.println(a);
14        } else if (b >= a && b >= c) {
15            System.out.println(b);
16        } else {
17            System.out.println(c);
18        }
19    }
20 }
```

OUTPUT

25

10
25
15

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new
6             Scanner(System.in);
7
8         int n = sc.nextInt();
9         int[] a = new int[n];
10
11         for (int i = 0; i < n; i++)
12             a[i] = sc.nextInt();
13
14         int x = sc.nextInt();
15         int l = 0, h = n - 1;
16
17         while (l <= h) {
18             int m = (l + h) / 2;
19             if(a[m] == x) {
20                 System.out.println(m);
21                 return;
22             }
23         }
24     }
25 }
```

OUTPUT

2

5
10 20 30 40 50
30

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new
6             Scanner(System.in);
7
8         int year = sc.nextInt();
9
10        if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
11            System.out.println("Leap Year");
12        } else {
13            System.out.println("Not a Leap Year");
14        }
15    }
16 }
```

OUTPUT

Leap Year

2024

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args){
5         Scanner sc = new
6             Scanner(System.in);
7
8         int n = sc.nextInt();
9         int a = 0, b = 1, c;
10
11         while (a <= n) {
12             System.out.println(a + " ");
13             c = a + b;
14             a = b;
15             b = c;
16         }
17     }
18 }
```

OUTPUT

```
0
1
1
2
3
```

20

QUESTIONS 31 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new
6             Scanner(System.in);
7
8         int a = sc.nextInt();
9         int b = sc.nextInt();
10
11         while (b != 0) {
12             int t = b;
13             b = a % b;
14             a = t;
15         }
16         System.out.println(a);
17     }
18 }
```

OUTPUT

6

INPUT

12
18

QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new
6             Scanner(System.in);
7         int n = sc.nextInt();
8
9         int i;
10        for (i = 2; i < n; i++) {
11            if (n % i == 0) {
12                System.out.println("Not Prime");
13                return;
14            }
15        }
16        if (n > 1)
17            System.out.println("Prime");
18        else
19            System.out.println("Not Prime");
20    }
21 }
```

OUTPUT

Prime

INPUT

7

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class SelectionSort {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter the number of elements: ");
8         int n = sc.nextInt();
9
10        int[] arr = new int[n];
11
12        System.out.println("Enter the array elements:");
13        for (int i = 0; i < n; i++) {
14            arr[i] = sc.nextInt();
15        }
16
17        for (int i = 0; i < n - 1; i++) {
18            int min = i;
19
20            for (int j = i + 1; j < n; j++) {
21                if (arr[j] < arr[min]) {
```

OUTPUT

```
Enter the number of elements: Enter the array elements:
Sorted array:
11 12 22 24 25
```

5
11
12
22
24
25

Table ▾

TABLE

Write a Java program to print the multiplication table of a given number using loops.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner s = new Scanner(System.in);
6         int n = s.nextInt();
7         for (int i = 1; i <= 10; i++)
8             System.out.println(n + " x " + i + " = " + (n * i));
9     }
10 }
```

OUTPUT

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
```

5

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

```
1 import java.util.Scanner;
2
3 public class Main {
4
5     static int fact(int n) {
6         if (n == 0 || n == 1)
7             return 1;
8         return n * fact(n - 1);
9     }
10    public static void main(String[] args) {
11        Scanner sc = new
12            Scanner(System.in);
13
14        int n = sc.nextInt();
15        System.out.println(fact(n));
16    }
17 }
```

OUTPUT

24

4

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

**
*

```
1 public class Pattern {  
2     public static void main(String[] args) {  
3  
4         for (int i = 5; i >= 1; i--) {  
5             for (int j = 1; j <= i; j++) {  
6                 System.out.print("*");  
7             }  
8             System.out.println();  
9         }  
10    }  
11 }  
12 }
```

OUTPUT

```
*****  
****  
***  
**  
*
```

Your INPUT goes here!

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class LCM {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int a = sc.nextInt();
8         int b = sc.nextInt();
9         int l = (a > b) ? a : b;
10
11         while (l % a != 0 || l % b != 0)
12             l++;
13
14         System.out.println(l);
15     }
16 }
```

OUTPUT

36

12
18

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class DtoB {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt(), b = 0, p = 1;
8
9         while (n > 0) {
10             b = b + (n % 2) * p;
11             n = n / 2;
12             p = p * 10;
13         }
14
15         System.out.println(b);
16     }
17 }
```

OUTPUT

1010

10

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class BtoD {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt(), d = 0, p = 1;
8
9         while (n > 0) {
10             d = d + (n % 10) * p;
11             p = p * 2;
12             n = n / 10;
13         }
14
15         System.out.println(d);
16     }
17 }
```

OUTPUT

10

1010

Sum ▾

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

```
1 import java.util.Scanner;
2 class SumDigits {
3     public static void main(String[] args) {
4         Scanner sc = new
5             Scanner(System.in);
6         int num = sc.nextInt();
7         int sum = 0;
8
9         while ( num > 0){
10             sum = sum + (num % 10);
11             num = num / 10;
12         }
13         System.out.println(sum);
14     }
15 }
```

OUTPUT

21

1234

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new
6             Scanner(System.in);
7
8         int n = sc.nextInt();
9         int product = 1;
10
11         while (n > 0){
12             int digit = n % 10;
13             product = product * digit;
14             n = n / 10;
15         }
16         System.out.println(product);
17     }
18 }
```

OUTPUT

120

456

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

```
1 import java.util.Scanner;  
2  
3 public class Power {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6  
7         int b = sc.nextInt();  
8         int e = sc.nextInt();  
9         int p = 1;  
10  
11         for (int i = 1; i <= e; i++)  
12             p = p * b;  
13  
14         System.out.println(p);  
15     }  
16 }
```

OUTPUT

32

2
5

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

```
1 import java.util.Scanner;
2
3 public class AreaCircle {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         double r = sc.nextDouble();
8         double a = 3.14 * r * r;
9
10        System.out.println(a);
11    }
12 }
```

OUTPUT

78.5

5

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter radius: ");
8         double r = sc.nextDouble();
9
10        double volume = (4.0 / 3.0) * 3.14 * r * r * r;
11
12        System.out.println("Volume of Sphere = " + volume);
13    }
14 }
```

OUTPUT

```
Enter radius: Volume of Sphere = 523.3333333333334
```

5

Palindrome ▾

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

```
1 import java.util.Scanner;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6         String s = sc.nextLine(), r = "";  
7  
8         for (int i = s.length() - 1; i >= 0; i--)  
9             r += s.charAt(i);  
10  
11         if (s.equals(r))  
12             System.out.println("Palindrome");  
13         else  
14             System.out.println("Not Palindrome");  
15     }  
16 }
```

OUTPUT

Palindrome

level

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt(), t = n, s = 0;
8
9         while (n > 0) {
10             int d = n % 10;
11             s += d * d * d;
12             n /= 10;
13         }
14
15         if (t == s)
16             System.out.println("Armstrong");
17         else
18             System.out.println("Not Armstrong");
19     }
20 }
```

OUTPUT

Armstrong

153

RT-num.Pattern ▾

**RT-
NUM.PATTERN**

Write a Java program to
print the following
pattern using loops:

1
22
333
4444
55555

```
1 public class Main {  
2     public static void main(String[] args) {  
3         for (int i = 1; i <= 5; i++) {  
4             for (int j = 1; j <= i; j++) {  
5                 System.out.print(i);  
6             }  
7             System.out.println();  
8         }  
9     }  
10 }
```

OUTPUT

```
1  
22  
333  
4444  
55555
```

Your INPUT goes
here!

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt(), s = 0;
8
9         for (int i = 1; i < n; i++)
10             if (n % i == 0)
11                 s += i;
12
13         if (s == n)
14             System.out.println("Perfect");
15         else
16             System.out.println("Not Perfect");
17     }
18 }
```

OUTPUT

Perfect

6

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6  
7         int n = sc.nextInt();  
8         int[] a = new int[n];  
9  
10        for (int i = 0; i < n; i++)  
11            a[i] = sc.nextInt();  
12  
13        int max = a[0];  
14  
15        for (int i = 1; i < n; i++)  
16            if (a[i] > max)  
17                max = a[i];  
18  
19        System.out.println(max);  
20    }  
21 }
```

OUTPUT

9

4
7 2 9 5

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int max = a[0], second = a[0];
14
15        for (int i = 1; i < n; i++) {
16            if (a[i] > max) {
17                second = max;
18                max = a[i];
19            } else if (a[i] > second && a[i] != max) {
20                second = a[i];
21            }
22        }
23    }
24 }
```

OUTPUT

25

5
10 25 8 40 15

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6  
7         int n = sc.nextInt();  
8         int[] a = new int[n];  
9  
10        for (int i = 0; i < n; i++)  
11            a[i] = sc.nextInt();  
12  
13        int min = a[0];  
14  
15        for (int i = 1; i < n; i++)  
16            if (a[i] < min)  
17                min = a[i];  
18  
19        System.out.println(min);  
20    }  
21 }
```

OUTPUT

8

5
10 25 8 40 15

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        int min = a[0], second = a[0];
14
15        for (int i = 1; i < n; i++) {
16            if (a[i] < min) {
17                second = min;
18                min = a[i];
19            } else if (a[i] < second && a[i] != min) {
20                second = a[i];
21            }
22        }
23    }
24 }
```

OUTPUT

10

5
10 25 8 40 15

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        for (int i = 0; i < n - 1; i++)
14            for (int j = 0; j < n - i - 1; j++)
15                if (a[j] > a[j + 1]) {
16                    int t = a[j];
17                    a[j] = a[j + 1];
18                    a[j + 1] = t;
19                }
20
21        for (int i = 0; i < n; i++)
```

OUTPUT

1 2 3 4 5

5
5 2 4 1 3

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int[] a = new int[n];
9
10        for (int i = 0; i < n; i++)
11            a[i] = sc.nextInt();
12
13        for (int i = 1; i < n; i++) {
14            int key = a[i], j = i - 1;
15
16            while (j >= 0 && a[j] > key) {
17                a[j + 1] = a[j];
18                j--;
19            }
20            a[j + 1] = key;
21        }
```

OUTPUT

1 2 3 4 5

5
5 2 4 1 3

RT-Pattern ▾

RT-PATTERN

Write a Java program to
print the following
pattern using loops:

```
**
***
****
*****
```

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args){
5         Scanner sc = new
6             Scanner(System.in);
7
8         int n = sc.nextInt();
9
10        for (int i = 1; i <= n; i++) {
11            for (int j = 1; j <= i; j++) {
12                System.out.print("**");
13            }
14            System.out.println();
15        }
16        sc.close();
17    }
18 }
```

OUTPUT

```
*
**
***
****
*****
```

5

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main (String[] args) {
4         Scanner sc = new
5             Scanner(System.in);
6
7         int n = sc.nextInt();
8         int fact = 1;
9         for (int i = 1; i <= n; i++)
10         {
11             fact *= i;
12         }
13         System.out.println(fact);
14     }
15 }
```

OUTPUT

120

INPUT

5